

29 Nov 2023

$G = (V, E, c)$ a ^{weighted} digraph, weights $c: E \rightarrow \mathbb{R}_{\geq 0}$ are "edge capacities"
 $s \in V$ called "source"
 $t \in V$ called "sink"

A flow $f: E \rightarrow \mathbb{R}_{\geq 0}$ such that
 f satisfies conservation of flow at all
 vertices except s & t , that is: $\forall v \in V \setminus \{s, t\}$

$$\sum_{u \in V} f(u \rightarrow v) = \sum_{w \in V} f(v \rightarrow w).$$

The value of the flow is

$$|f| = \sum_{u \in V} f(s \rightarrow u) - \underbrace{\sum_{w \in V} f(w \rightarrow s)}_{\text{most of the time, this is } = 0.}$$

Read: The value of the flow is
 equal to the net amount leaving the source.

Claim: The value of the flow is equal to the net amount entering the sink.

$$|f| = \sum_{u \in V} f(t \rightarrow u) - \sum_{w \in V} f(w \rightarrow t)$$

Proof: $|f| = - \sum_{u \in V} f(s \rightarrow u) + \sum_{w \in V} f(w \rightarrow s)$ by def'n.

$$\text{Let } \partial f(v) := \sum_{w \in V} f(w \rightarrow v) - \sum_{u \in V} f(v \rightarrow u)$$

Then, $|f| = \partial f(s)$ and

$$\begin{aligned} \sum_{v \in V} \partial f(v) &= \sum_{v \in V \setminus \{s, t\}} 0 + \partial f(s) + \partial f(t) \\ &= 0 + \partial f(s) + \partial f(t) \end{aligned}$$

For each edge $u \rightarrow v$, it appears $2x$ in $\sum_{v \in V} \partial f(v)$, once as $+f(w \rightarrow v)$ in $\partial f(v)$ and once as $-f(w \rightarrow v)$ in $\partial f(w)$.

$$\therefore \sum_{v \in V} \partial f(v) = \sum_{\substack{e \in E \\ (w \rightarrow v)}} +f(w \rightarrow v) - f(w \rightarrow v) = \sum 0 = 0$$

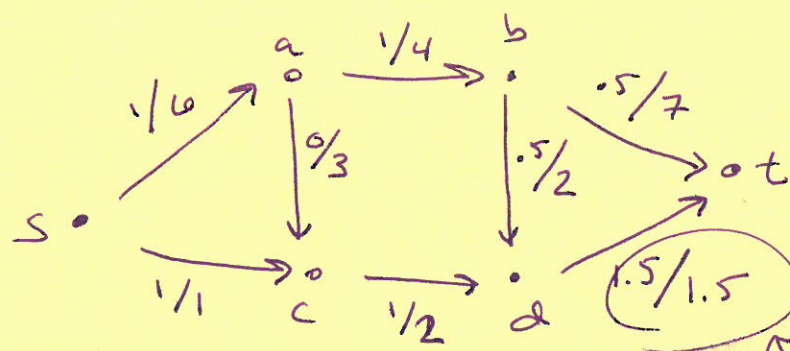
Putting these together, we get

$$0 = 0 + \partial f(s) + \partial f(t) \Rightarrow \partial f(s) = -\partial f(t)$$

$$\text{Thus, } |f| = \partial f(s) = -\partial f(t) = \sum_{u \in V} f(t \rightarrow u) - \sum_{w \in V} f(w \rightarrow t) \quad \square$$

(2)

Suppose we have a flow graph & a flow:
 $G = (V, E, c)$ $f: E \rightarrow \mathbb{R}$



$f(e) = c(e)$
 the edge is "saturated"

The residual graph

is the graph $G = (V, V \times V \setminus \Delta, r)$
 $\uparrow$
self-edges

where $r: V \times V \setminus \Delta \rightarrow \mathbb{R}$ is defined by

$$r(e) = \begin{cases} c(e) - f(e), & e \in E \\ -f(e^{-1}), & e^{-1} \in E \\ 0, & \text{otherwise} \end{cases}$$

$e = (u, v)$
 $e^{-1} = (v, u)$

$\uparrow$
 $r(e)$ tells us how much we can change the flow by on edge e .

$\rightarrow$ i.e. w/out explicitly constructing the residual graph.

Question How can I implicitly check to see if a path exists from s to t in the residual graph?

Ford Fulkerson

MAX FLOW ($G = (V, E, c), s, t$)

- 1: ~~path~~ find a path from s to t in G
- 2: $f \leftarrow$ a ~~fun~~ $f: V \times V \setminus \Delta \rightarrow \mathbb{R}$
initialized to zero everywhere (trivial flow)
- 3: while path exists
- 4: | augment flow f along path
- 5: | path $\leftarrow$ a path in the residual graph for (G, f) using only non zero weight edges
- 6: endwhile
- 7: return f

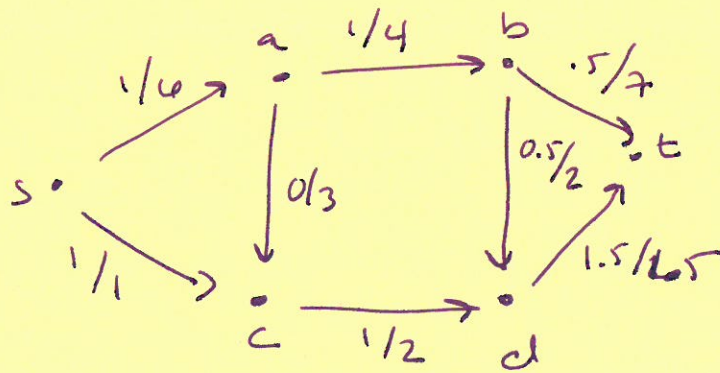
Augmenting a path

- $\forall e \in \text{path}$ st. $e \in E$
 $f(e) = f(e) + \text{path min}$
- $\forall e \in \text{path}$ st. $e \notin E \Rightarrow e' \in E$
 $f(e') = f(e') - \text{path min}$

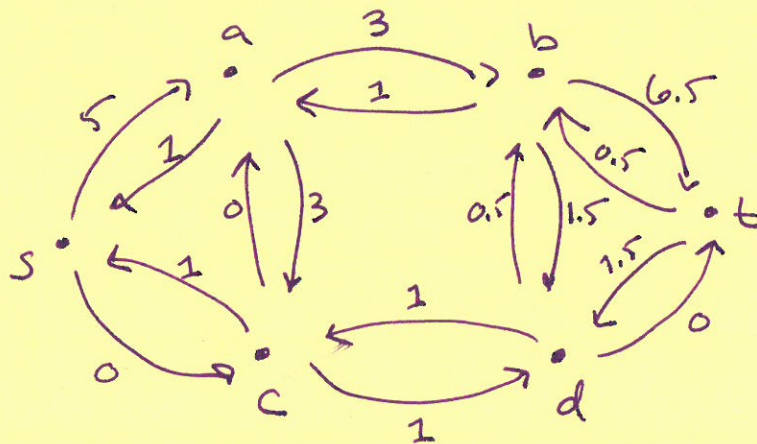
Questions

1. Find an example that requires undoing flow
2. Walk through an example of this algorithm.
3. Why does ~~a~~ path in ~~any~~ ^{resid} graph
 $\Rightarrow$ that f is maximized?

What is the residual graph for:



Residual is:

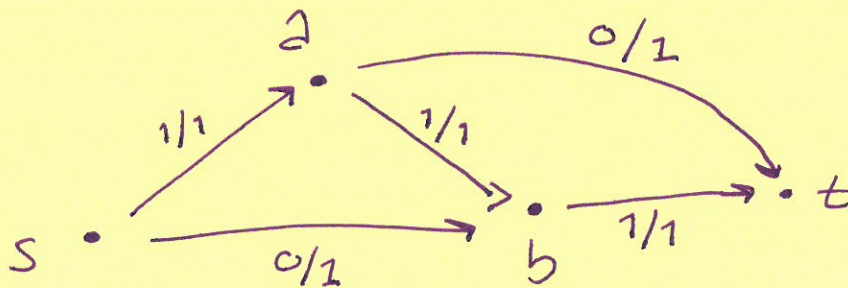


Paths using nonzero weight edges from s to t :

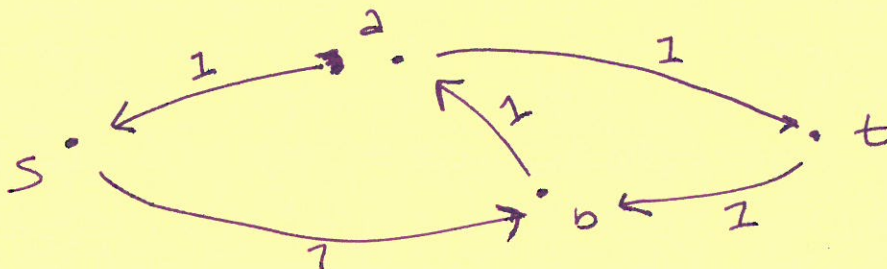
$\{s, a, b, t\}$

$\{s, a, c, d, b, t\}$

Q1:



$|f| = 1$



$\{s, b, t\}$